



Impact Crater

The AI-Powered Reel and Video Generator.
Simply dump your media, describe your
desired output, and save hours of editing.

Product & Architecture Overview

Impact Crater

AI Based Video and Reel Creator for Your Media



00:00:07:48

00:00:05:12

00:00:10:24

00:08:15:36

THE DIGITAL GRAVEYARD



CORE BLOCKERS

Selection Fatigue

The overwhelming task of picking the best shots.

INPUT: 5,660+ CLIPS / PROCESSED: 0
STATUS: PENDING REVIEW

Sequencing

The craft of structuring a coherent story.

FLOW STATE: DISRUPTED / NARRATIVE STRUCTURE: UNDEFINED / TIMELINE: 00:00:00:00

Syncing

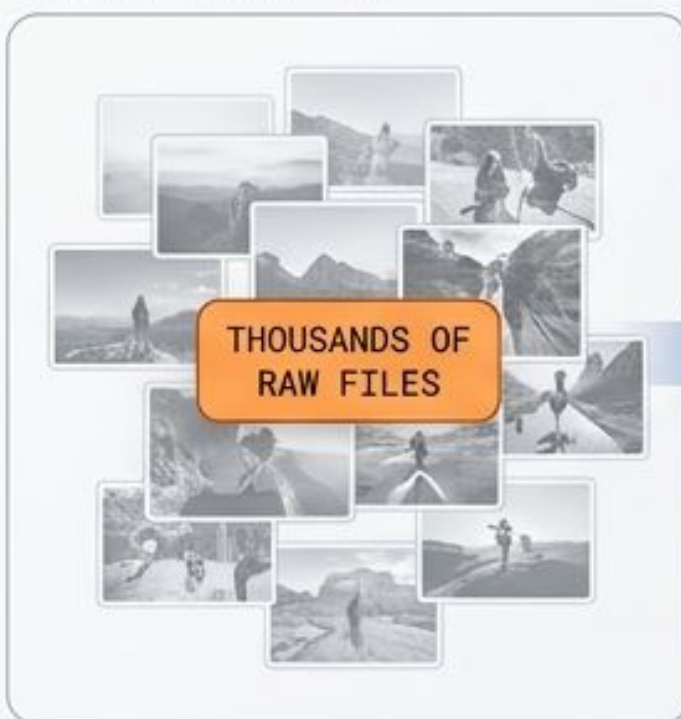
The technical hurdle of matching cuts to music.

SYNC OFFSET: +0.45s / BEAT MATCH: FAILED /
AUDIT STATUS: MISALIGNMENT DETECTED

THE PROBLEM

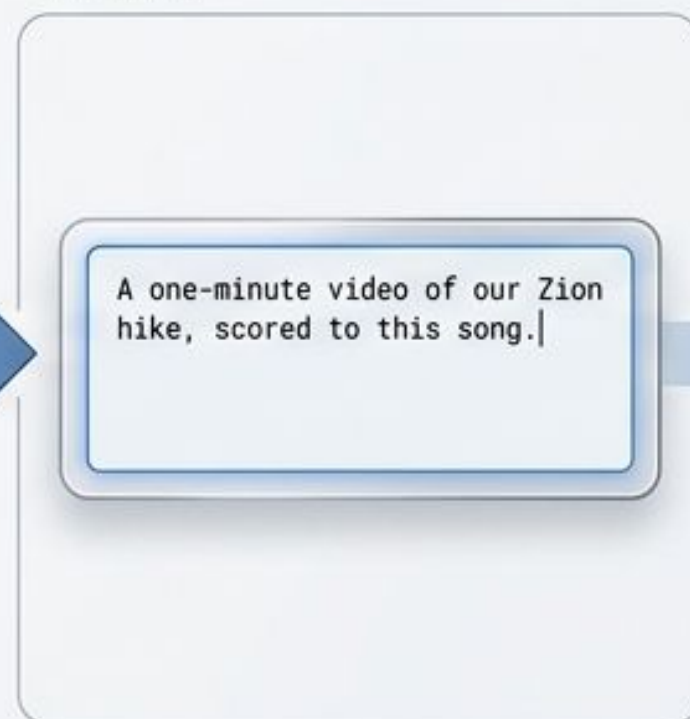
We capture thousands of photos and videos, but rarely create anything with them. Curation is overwhelming. Editing is tedious. Matching footage to music is a specialized craft.

DROP FOLDER



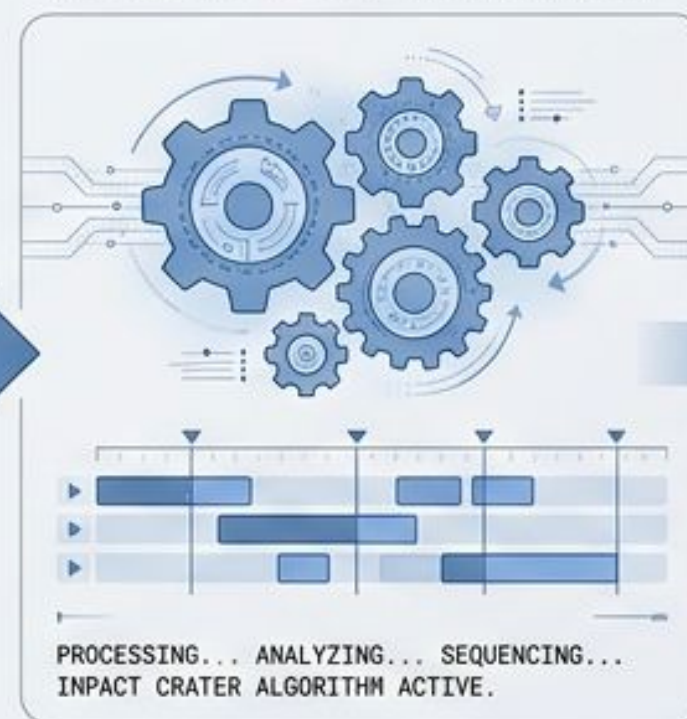
Provide raw media.

BRIEF



Write a plain-language prompt (e.g., A one-minute video of our Zion hike, scored to this song).

AUTONOMOUS CURATION



Impact Crater analyzes, selects, and sequences.

APPROVE



Nothing publishes without human consent.

PLAIN LANGUAGE IN, FINISHED VIDEO OUT



Drop in raw media

+

> A one-minute video of
our Zion hike, building
to the canyon overlook,
scored to this song.



The system analyzes the media, picks the best moments, fits them to music, and renders the result – automatically.

Story Videos tailored to the mood

Standard Mode

Your music plays gracefully under a narrative cut of the best moments.



Music-Video Mode

Cuts are synchronized precisely to the beat grid of the song.



Steer the edit: Map 'the climb' to the build-up, and 'the summit' to the chorus using plain language.

Creative Outputs & Roadmap

Available Today

Standard Story Video

Concept: A narrative cut of best moments over background music.

Capabilities: Forward-in-time flow, landscape 16:9, YouTube-publishable.

Available Today

Music-Video Mode

Concept: Dynamic cuts synchronized perfectly to the song's beat grid.

Capabilities: Plain-language steering (e.g., build-up over the climb, chorus on the summit), auto-generated cut points.

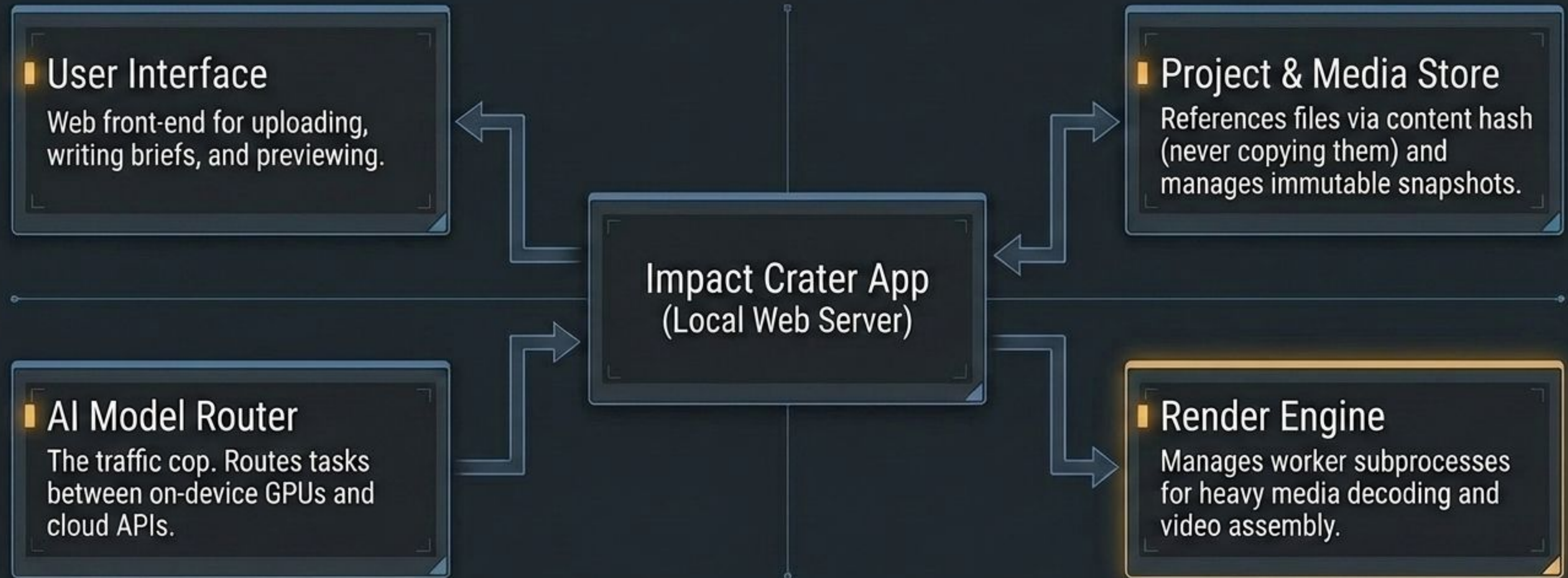
Roadmap

The Trip Package

Concept: An entire multi-day trip dropped at once.

Capabilities: Automatically plans a coordinated set of assets (per-location reels, multi-photo collages, overall trip film).

It is a Self-hosted AI-based Media-to-Video Creation Engine



Installed and launched as a desktop-class local app. Designed so hosted-service mode is a future configuration, not a rewrite.

Architectural Principles



Self-Hosted First

Runs as a local desktop app. Media stays on your machine by default.



Hardware-Adaptive

Automatically balances on-device vs. cloud AI based on local GPU constraints and budget.



Preview, Then Approve

A hard gate. Nothing posts publicly without explicit human consent.



Cost-Conscious

Heavy use of cheap/fast models; high-reasoning AI used sparingly. Re-runs are nearly free due to caching.



Transparent

Every decision is inspectable (what was kept, dropped, why, and cost).



Privacy-Aware

Location data and faces feature explicit, default-on protections.

PHASE 1: PREPARE

PHASE 2: ANALYZE

PHASE 3: EXECUTE

1. Ingest & Understand

Every file is thumbnailed, content-hashed, and videos are automatically split into discrete scenes.

STATUS:
INDEXING
FILES: 1.4K
SCENES: 3.2K

2. Per-Asset Analysis

Fast AI generates a one-line caption, a technical quality score (focus, exposure, motion blur), relevance score, and a visual fingerprint embedding.

CAPTION:
"Sunset over mountain range"
QUALITY SCORE:
8.5/10
RELEVANCE:
HIGH
EMBEDDING:
512d

3. Rich Metadata Extraction

Mid-tier AI identifies shot type (wide, close-up), scenery, lighting, and assigns an intrinsically special shot score.

SHOT TYPE:
WIDE
SCENERY:
MOUNTAINS
LIGHTING:
GOLDEN HOUR
SPECIAL SCORE:
9.2/10

Phase 1: Prepare the raw material

Ingest & Fingerprint

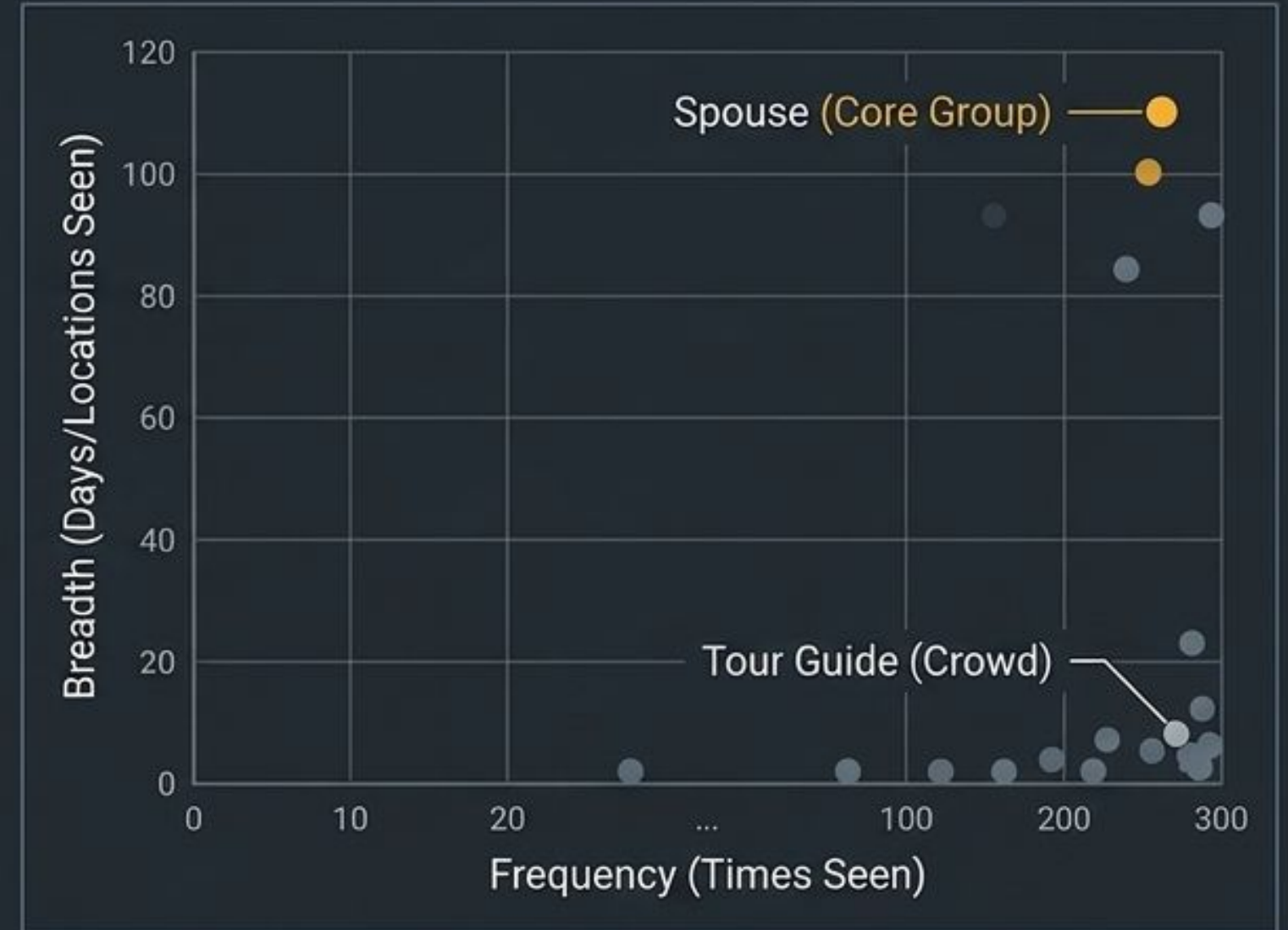
Unique content hashes and scene splitting.

Per-Asset Analysis

Fast AI extracts one-line captions, technical scores, and embeddings.

Rich Metadata

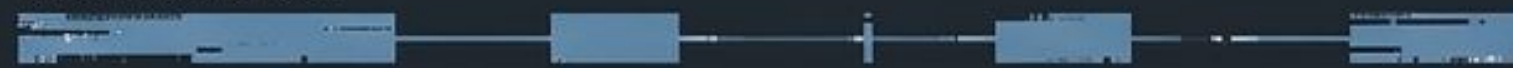
Extracting lighting, shot types, expressions, and safety ratings.



The system learns who the trip is actually about, ensuring the final video prioritizes the right people.

Capture-Time Reconciliation

Embedded EXIF



Filename Patterns (IMG/Screenshot)



File Modified Dates



Master Chronology

YYY-MM-DD_HH:MM:SS

Two-Stage Deduplication



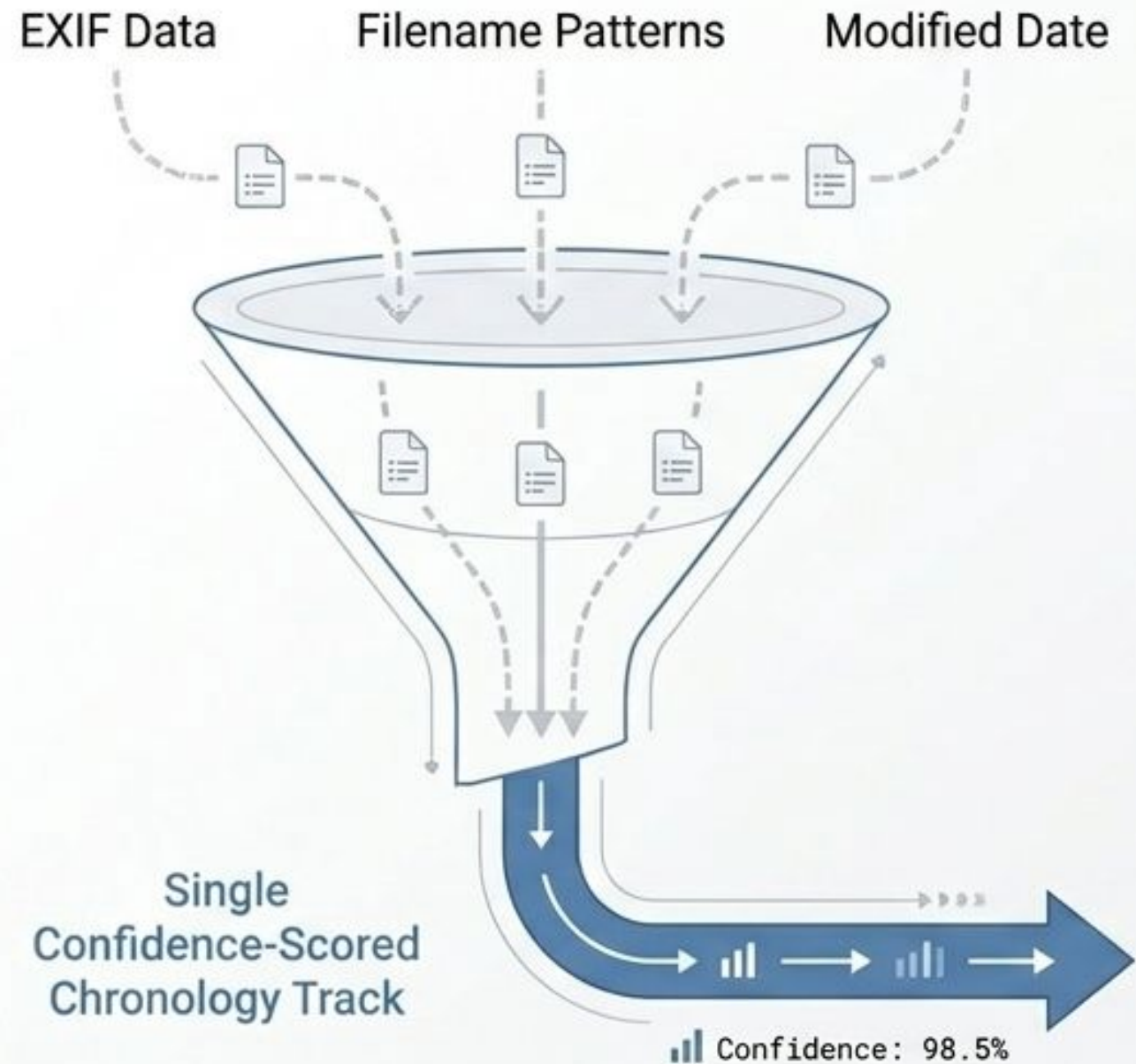
Stage 1: Pixel-Level Hashing
(Catches exact digital duplicates)

Stage 2: Visual Fingerprint + Time Window
(Catches rapid-fire retakes from slightly different angles)

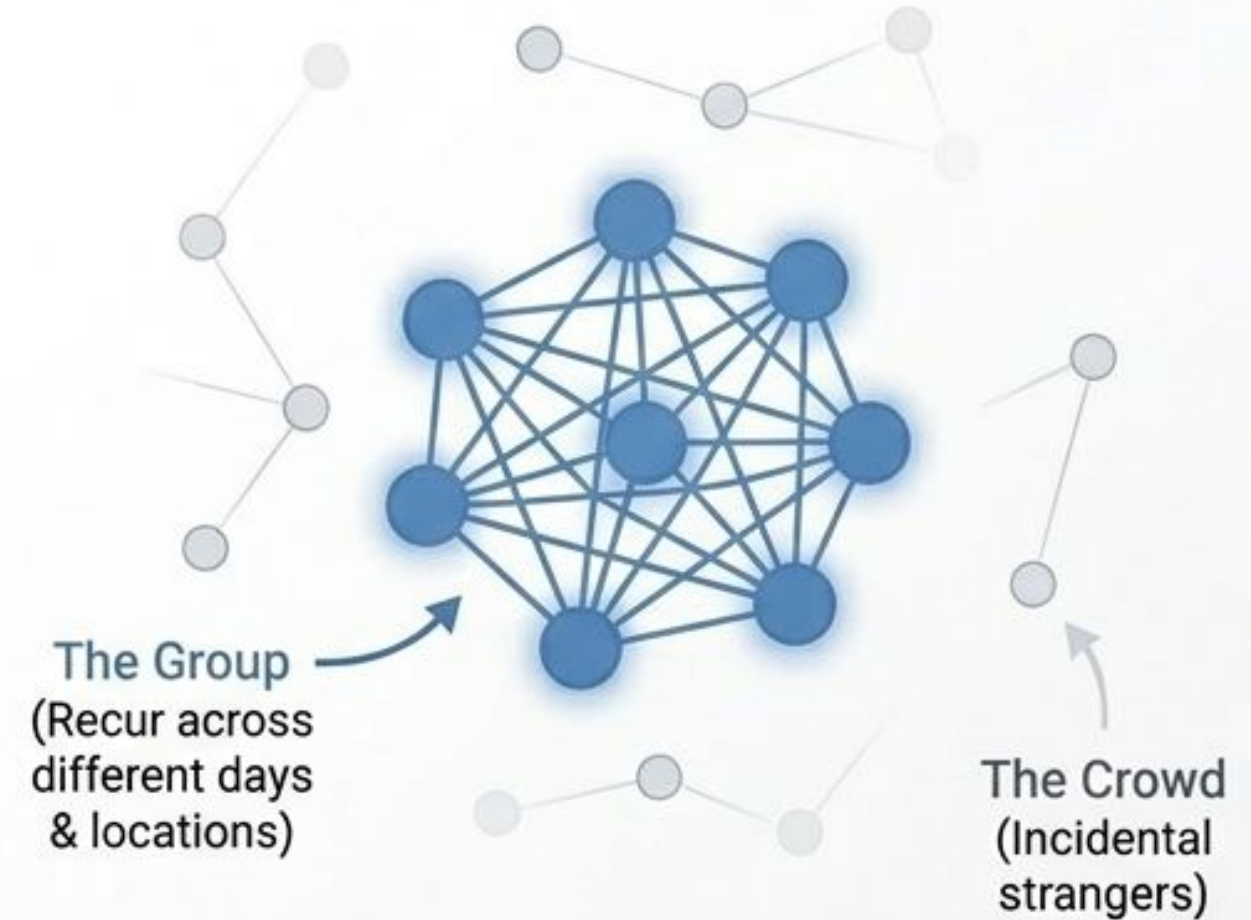


Only the single best shot
of that moment survives.

Capture-time Reconciliation



Recurrence-Breadth Group Detection



Differentiates core travel companions from background strangers based on breadth of recurrence, not just total screen time.

1. Smart Pre-filter

A fast rule-based cull. Drops unsafe frames, below-floor quality, and thins out over-photographed locations. Ranks by quality, brief-relevance, and diversity.

2. Narrative Judgment

A high-reasoning AI reviews the candidate set as a whole. Produces an ordered story, assigning roles (opener, build, peak, closer).

3. Timeline Planning

Converts the story into concrete per-clip durations, smart-cropping, and snaps cut points to a song's beat grid.

4. Second-Guess Review

An AI sanity-check pass that surfaces high-confidence improvements for user confirmation.

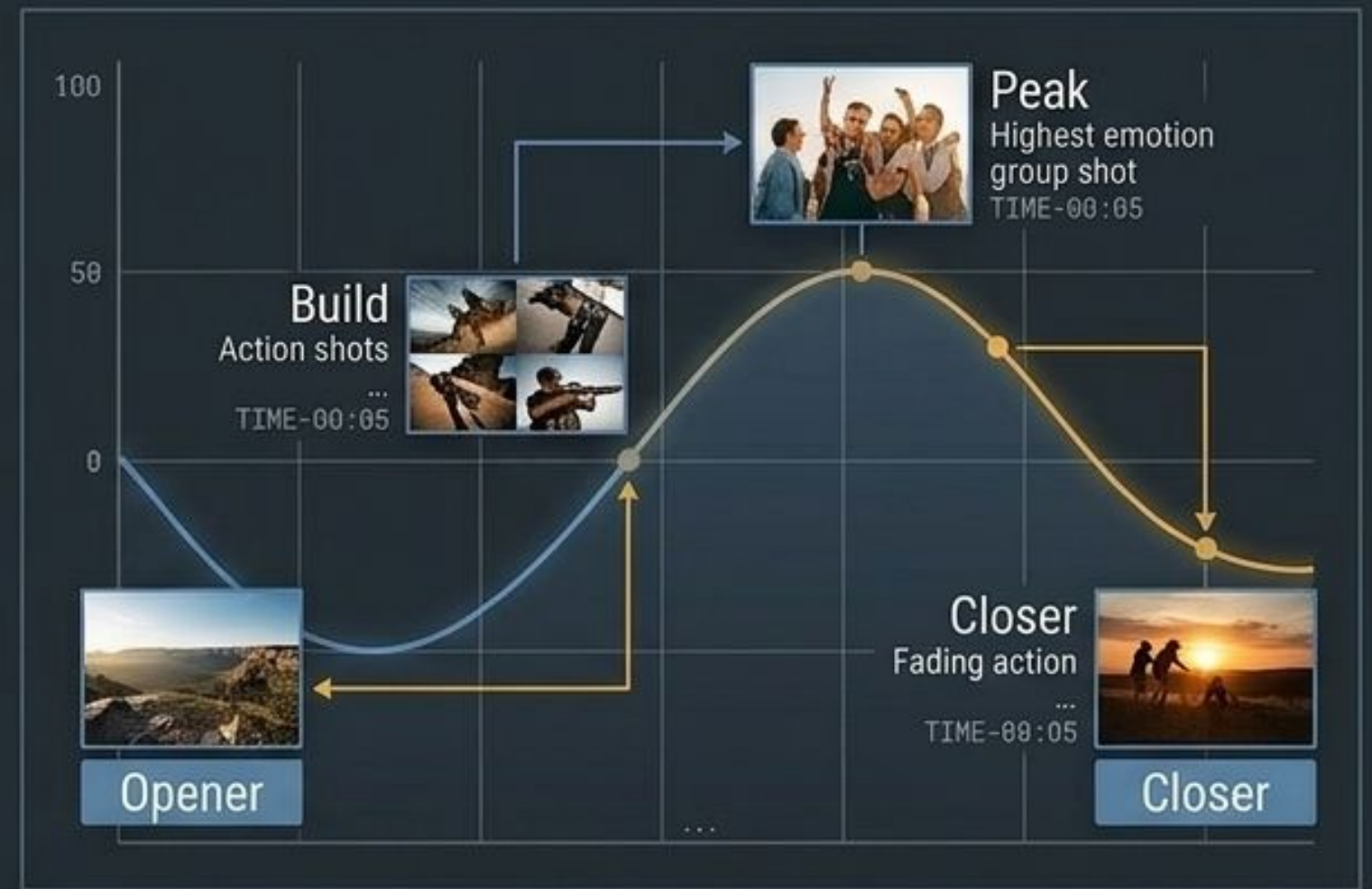
Phase 2: Narrative-Arc-As-Judge

The Flawed Standard



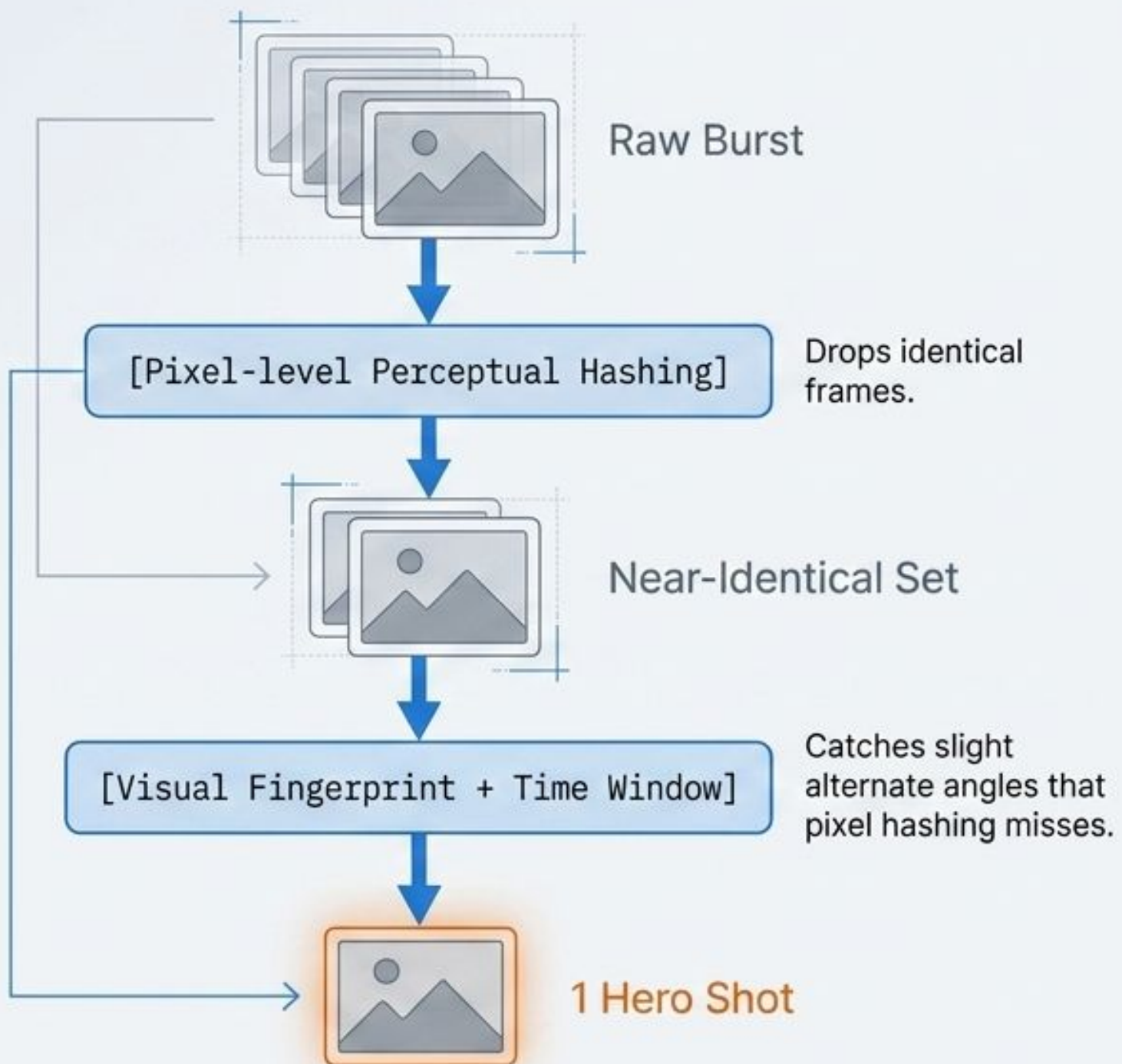
Greedy Selection: Grading photos individually results in monotonous highlight reels with zero narrative pacing.

The Impact Crater Way



A high-reasoning AI model composes the entire ordered story in a single pass, considering metadata, shot type, and chronology to build a coherent narrative.

Two-Stage Duplicate Suppression



Evaluating the Whole, Not the Parts



Narrative-Arc-As-Judge

Instead of greedy per-shot selection, the high-reasoning AI composes the entire ordered story in a single pass to ensure true narrative coherence.

RENDER_ENGINE V1.2 | AUDIO_NORM_dBFS | FRAME_ASSEMBLY

1. Render with Music

Video engine assembles clips, normalizes audio to broadcast loudness, applies smooth fades, and muxes the final file.

Smartly re-judges the story or re-analyzes specific shots using cached data for near-instant updates.



2. The Refinement Loop

3. The Approval Gate

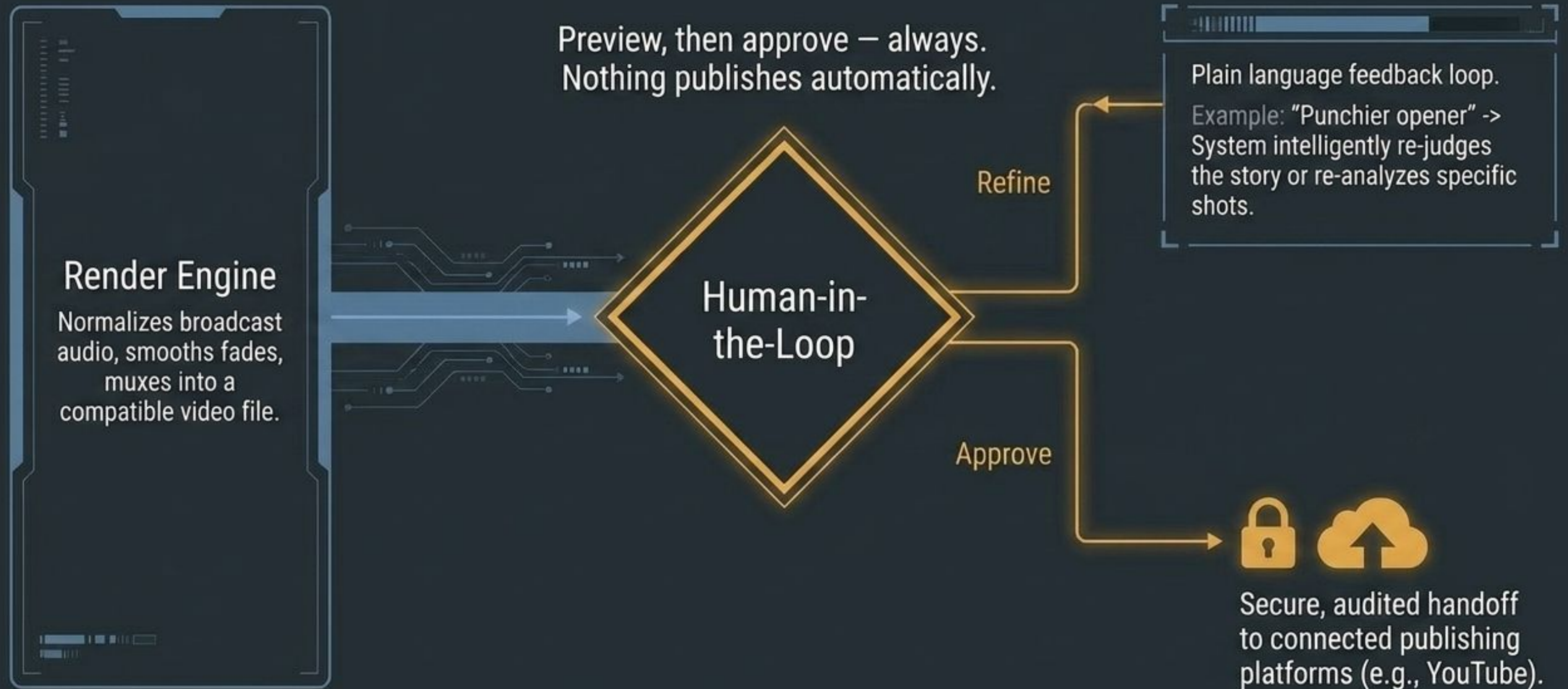
You watch the result. Two choices: Refine or Approve. Nothing reaches YouTube without the user turning the key and verifying visibility settings.



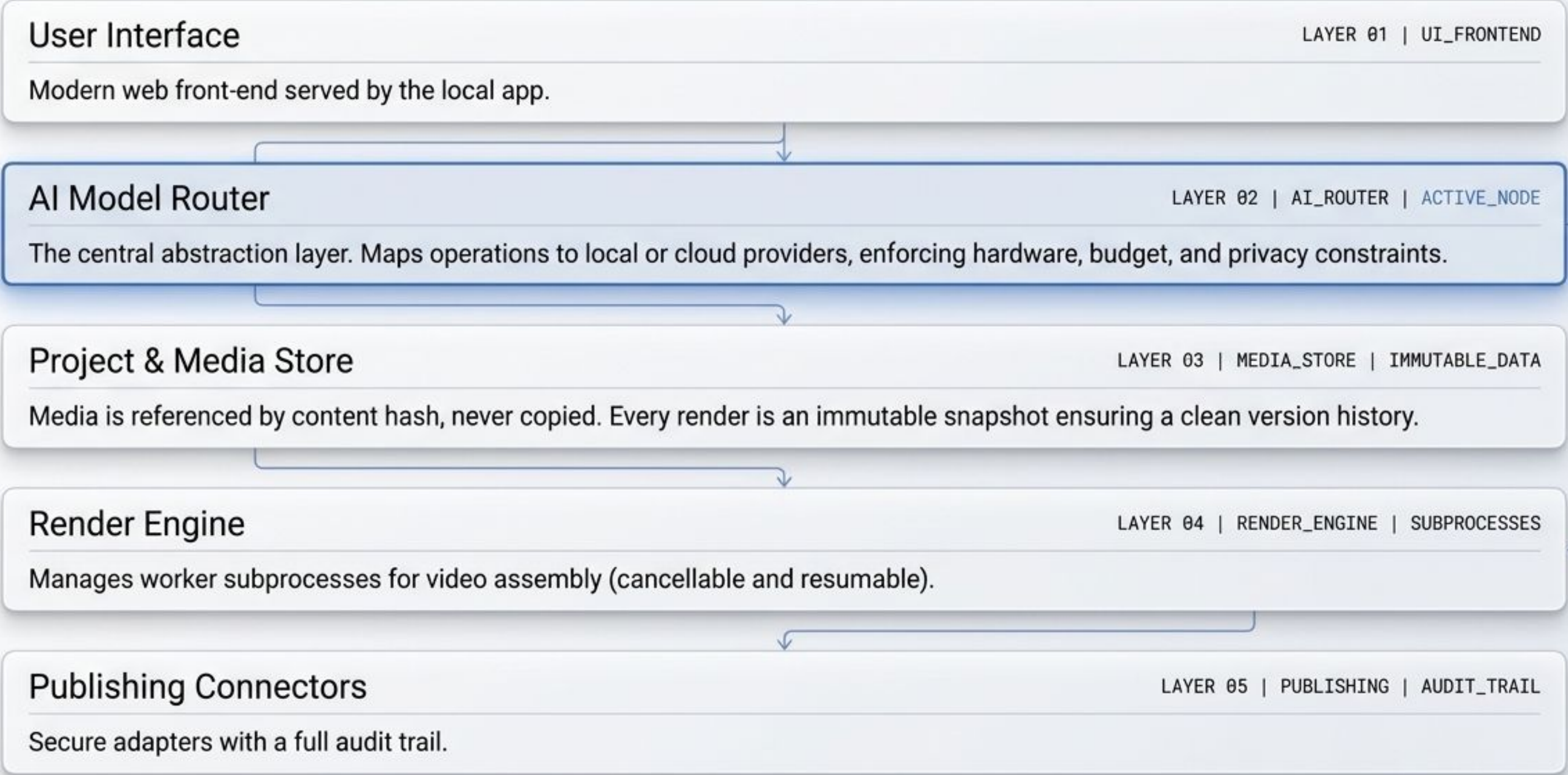
RENDER_ENGINE V1.2 | AUDIO_NORM_dBFS | FRAME_ASSEMBLY

THE APPROVAL GATE

Phase 3: Produce, Preview, and Publish



System Architecture



AI Model Tiering Strategy

	Col 1 Tier 1 (Small/Fast)	Col 2 Tier 2 (Mid)	Tier 3 (Large/High-Reasoning)
Primary Tasks	Captions, quick scores, visual embeddings.	Rich metadata, orchestration reasoning.	Composing the final narrative.
Execution Volume	High (Every frame/photo) 	Moderate (Filtered candidate set) 	Rare (Used once per job)
Cost Profile	Ultra-cheap.	Efficient.	Expensive but contained.

Interchangeability: Every model sits behind a unified interface; local on-device models can stand in for cloud models based on hardware constraints.

Hardware-Adaptive Routing & Privacy

Scenario A: Lightweight Laptop (No GPU)

LAYER 06 | LOCAL_PREP_CLOUD_ROUTE

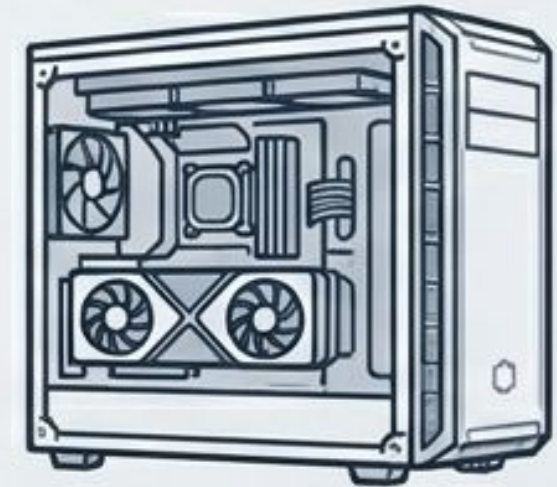


Routing: Leans heavily on cloud AI services.
Routing: Leans heavily on cloud AI services.

Privacy Protocol: Embedded location data stripped.
Faces blurred locally before media leaves the device.

Scenario B: High-VRAM Workstation

LAYER 06 | ON_DEVICE_ROUTE_SECURE



Routing: Models are capped at consumer-GPU sizes
to run entirely on-device.

Privacy Protocol: 100% localized analysis.
Identifiable data never leaves the machine.

Cost Control Panel

UI features hard daily spending caps and live spend tracking by provider.

CAP_LIMIT: \$10.00 / DAY

CURRENT_SPEND: \$7.25

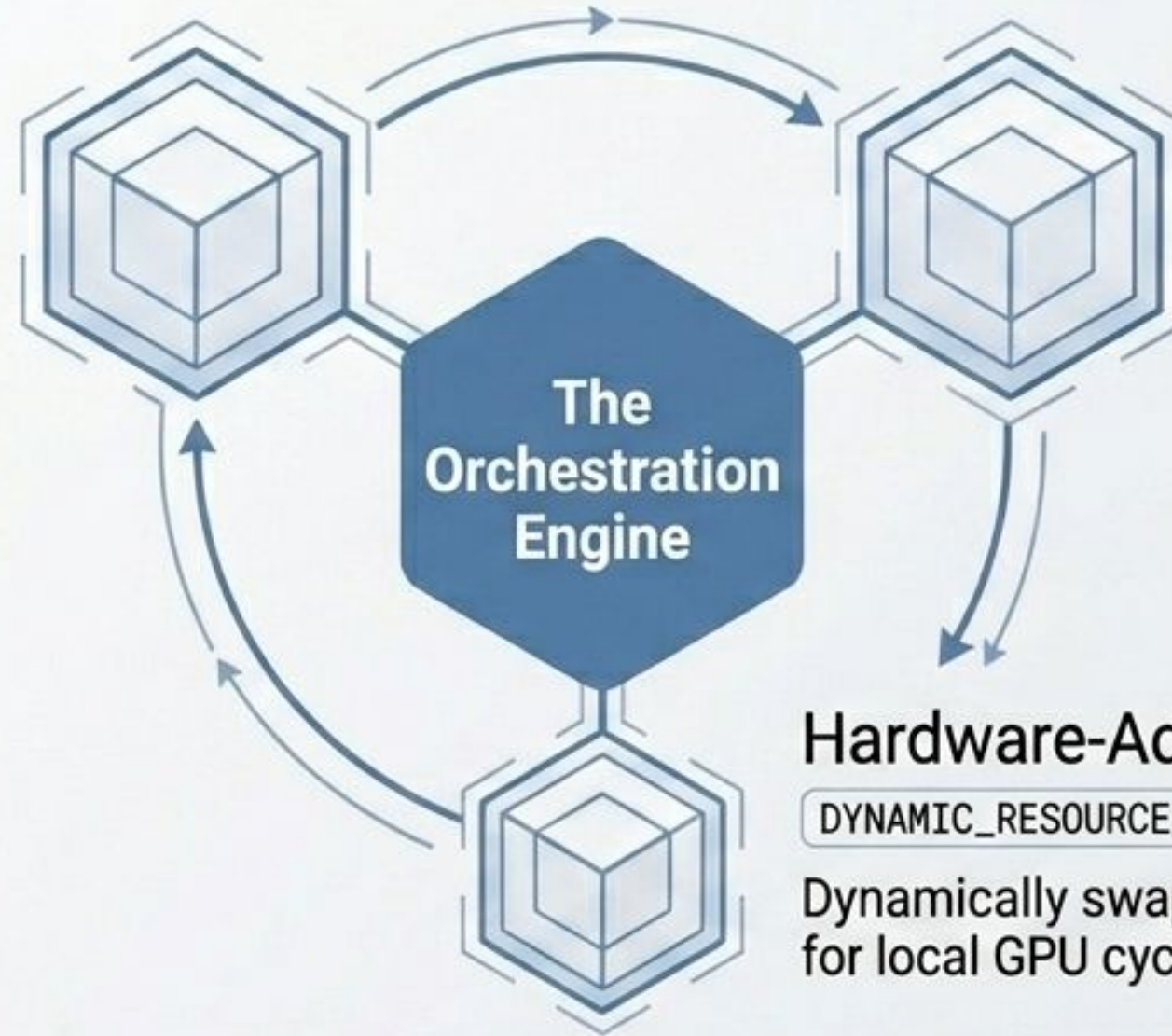


Synthesis: The Orchestration Engine

Content-Addressed Caching

CACHING_PROTOCOL

Cached by content hash + model + exact operation. Identical photos are never analyzed twice.



Cheap-First Analysis

PRE_ANALYSIS_OPTIMIZATION

Heavy per-item analysis runs on downscaled renditions (10x less data), caching to the full-res original.

Hardware-Adaptive Routing

DYNAMIC_RESOURCE_ALLOCATION

Dynamically swapping cloud APIs for local GPU cycles.

Synthesis: Impact Crater isn't just an AI wrapper; it is an **ultra-efficient orchestrator** designed to stretch computational dollars. Refining a video or re-running a job isn't starting over—it's **nearly free**.

Hardware-adaptive AI routing

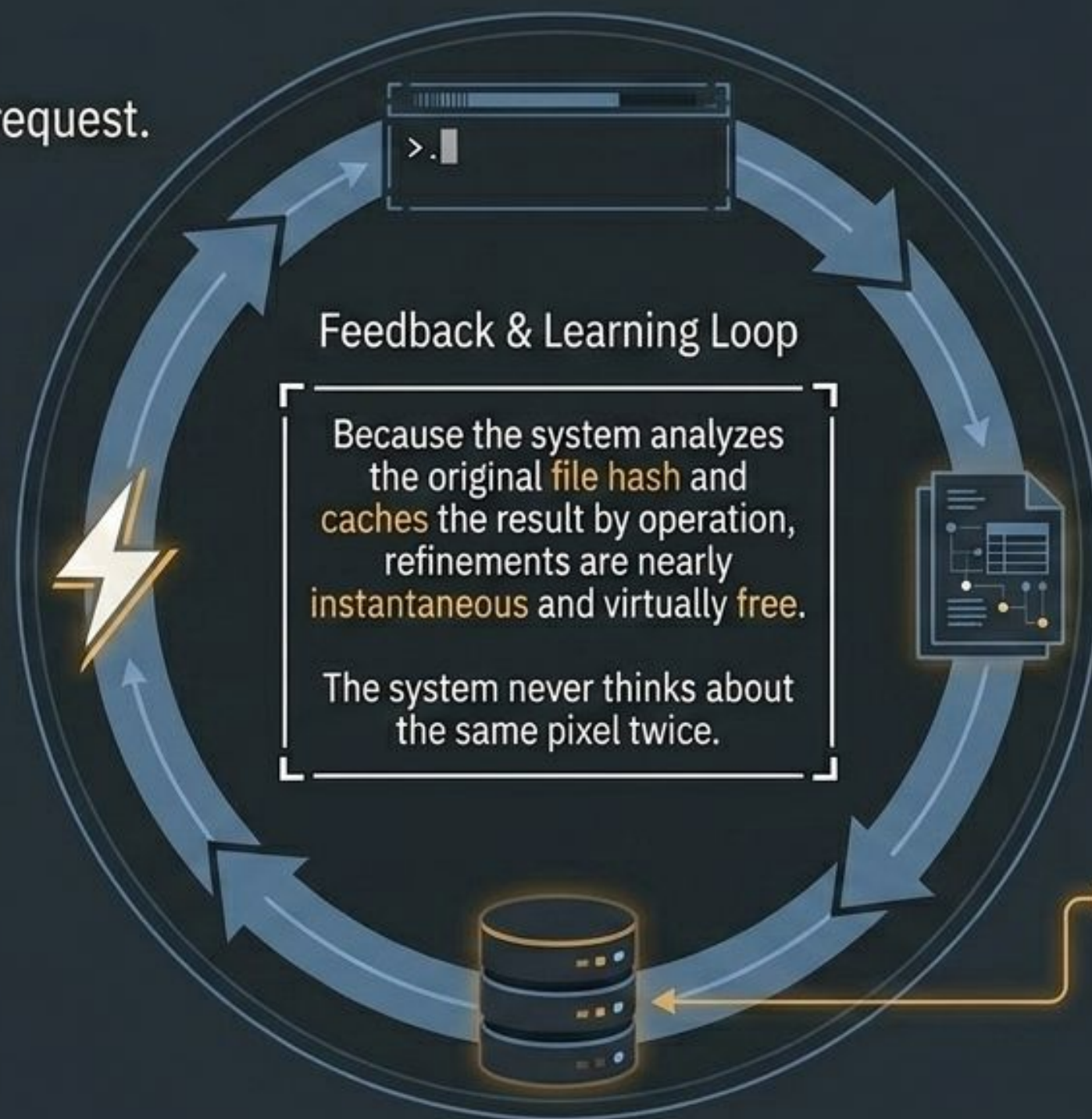
[MODEL TIER]	[TASKS ASSIGNED]	[PROCESSING LOCATION]	[COST & SPEED PROFILE]
Small / Fast	Captions, quick scores, visual embeddings.	Local or Cloud (Hardware dependent).	High volume, fractions of a cent, ultra-fast.
Mid	Rich metadata extraction, orchestration logic.	Local or Cloud.	Balanced workhorse.
Large / High-Reasoning	Narrative composition and human-feedback translation.	Cloud API.	Expensive, used sparingly (once per job).

The router balances on-device GPUs and cloud APIs dynamically based on your hardware capabilities and hard spending limits.

The speed secret: Content-addressed caching

Plain language refinement request.

Instant re-planning.
Only the large narrative
model runs again.



Feedback & Learning Loop

Because the system analyzes the original **file hash** and **caches** the result by operation, refinements are nearly **instantaneous** and virtually **free**.

The system never thinks about the same pixel twice.

Immutable Snapshot Check.
System pulls the exact plan of
the previous render.

0x8F92A4E1C...

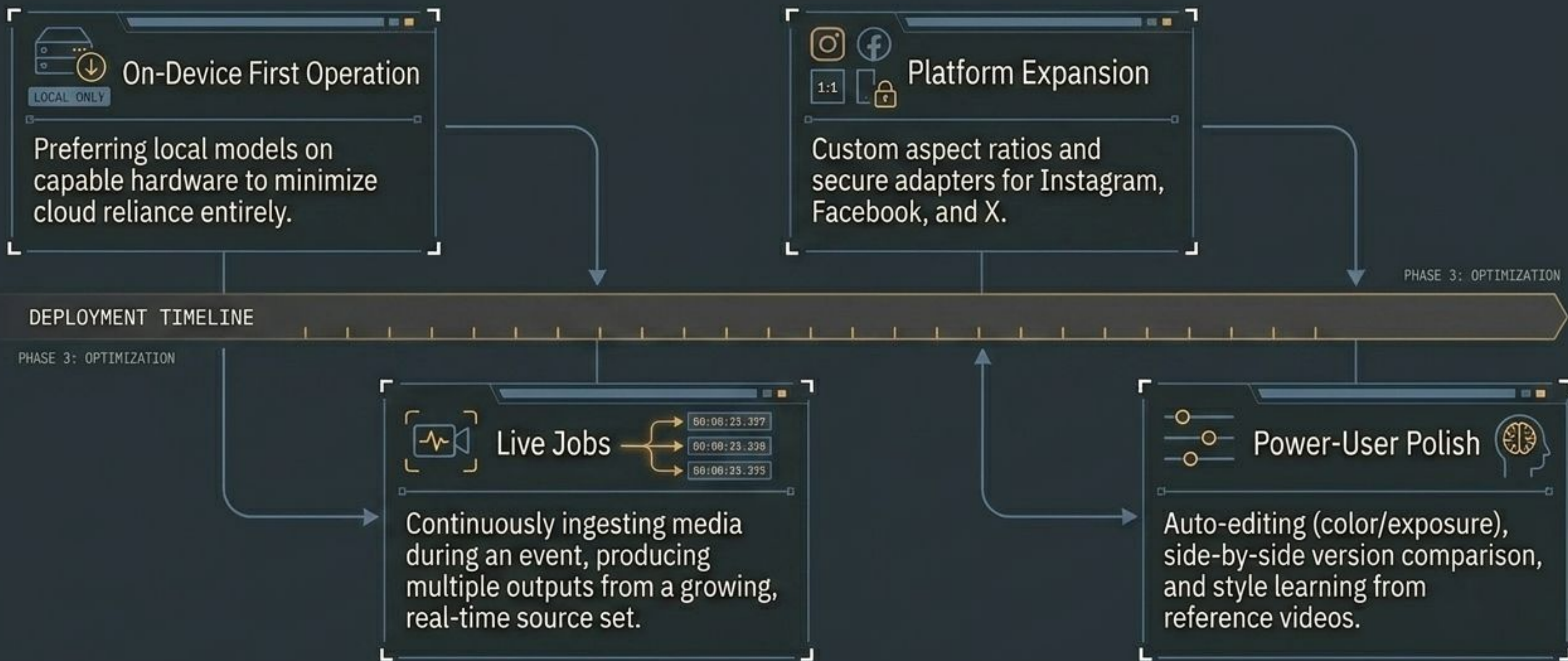


Content-Addressed Cache.
The raw photo isn't re-processed.

Trust and control by default



Near-term evolution



The Vision: The Trip Package

14-Day Europe Trip
(5,000 raw items)

Place Videos



Paris Highlights
(Full Cut)



Rome Walkthrough
(Full Cut)

Reels



Short, punchy vertical clips mapped
to standout, high-energy moments.

The Master Film



A comprehensive, multi-chapter
chronological trip documentary.

A completely coordinated package, planned from a single analysis pass, delivered behind one unified approval surface.

Available to host today

Command the edit.

Drop in media. Describe your vision.

Full prepare-plan-produce **pipeline** is live.

Story videos available for
YouTube publishing.

The License



Open source under the
Business Source License (BSL 1.1).

Free to self-host for personal, family, or internal team use.
Converts to a fully permissive license on its change date.

`impactcrater.app // github.com/impactcrater`

Product Roadmap

Near-Term Fixes

Live Jobs:

Ingesting media continuously during an event.

Style Learning:

Deriving pacing and look preferences from reference videos.

Auto Editing:

Per-shot color and contrast improvements.

Horizon Innovations

The Trip Package:

True multi-asset orchestration from a single trip dump.

AI Erase & Score:

AI-generated music & automatic background crowd removal.

Hosted Mode:

Multi-tenant service using the exact same underlying architecture.



Open Source Architecture

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The Terms

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The Philosophy

Self-hosted by default. Telemetry is local-only. Your media stays yours. The design ensures a future hosted-service is just a configuration change, not a re-architecture.